COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	8/28/2026	**DATE SUBMITTED:**	8/31/2026
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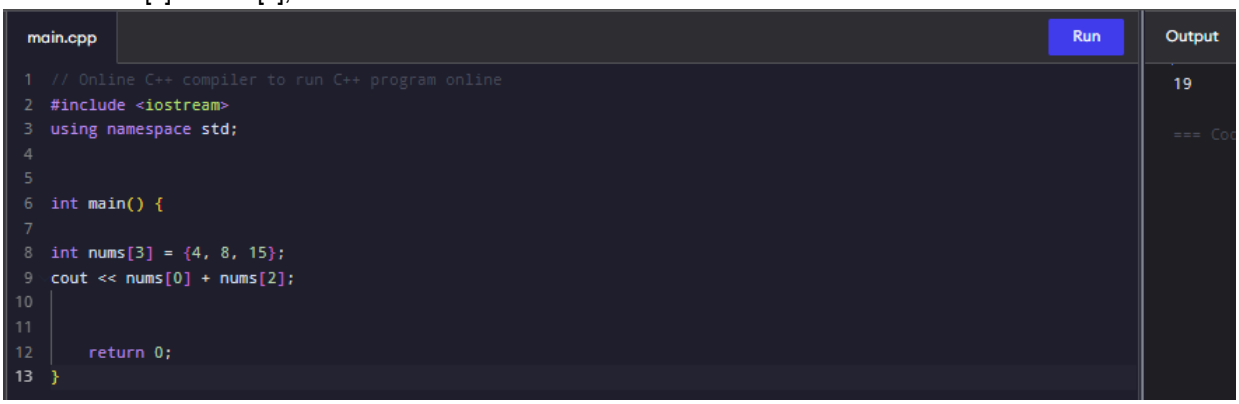
ACTIVITY NO. 4.1 Arrays

1. PROCEDURE OUTPUT

What procedures did you perform?

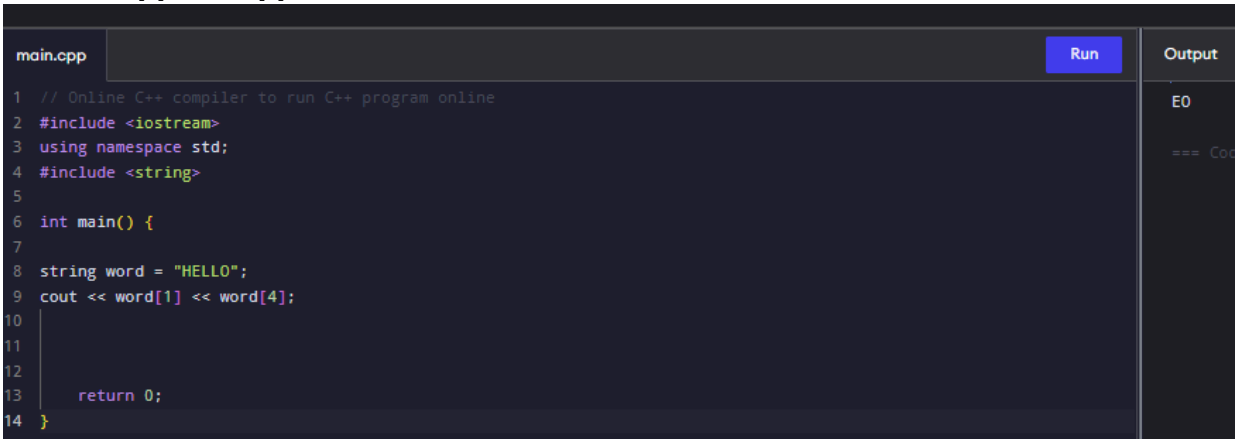
Write what happens in each snippet:

1. int nums[3] = {4, 8, 15};
cout << nums[0] + nums[2];



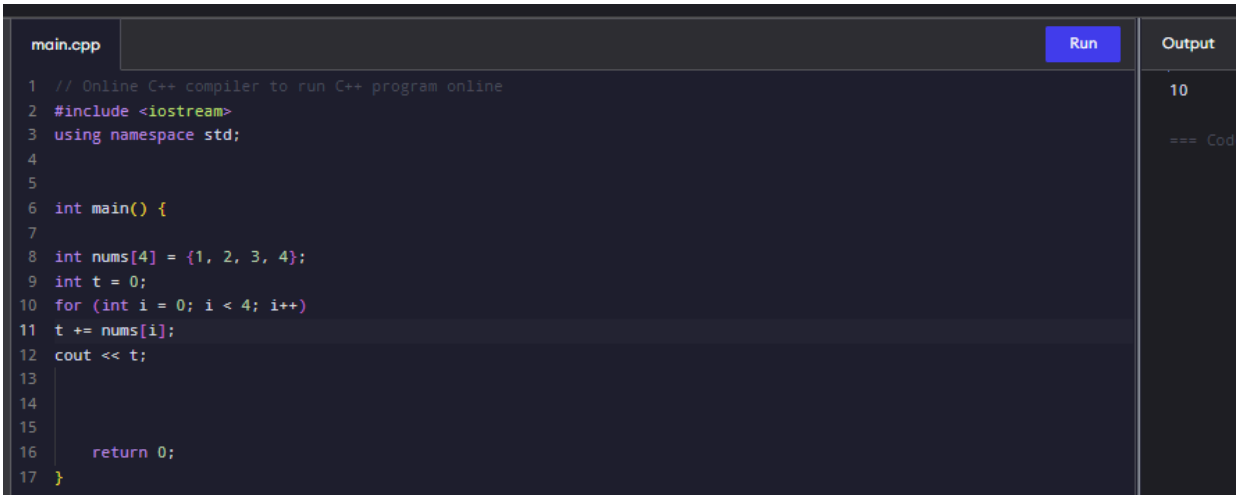
Upon uploading the given code, the program first located the indicated positions which are 0 and 2 that correspond to 4 and 15 where both numbers were combined using the operation addition resulting in the output of 19.

2. string word = "HELLO";
cout << word[1] << word[4];



The program made use of the string function wherein the data in the array are actually each letters that make up the word “HELLO”, wherein the count still retains, 0 is letter H while 4 is the letter 0.

3. int nums[4] = {1, 2, 3, 4};
int t = 0;
for (int i = 0; i < 4; i++) t += nums[i];
cout << t;



Utilizing the for loop for this number, the program uses a loop so that the program checks all the data from the array wherein it gets combined until all the numbers inside the array are added together.

4. int t[2][2] = {{1, 2}, {3, 4}};
cout << t[1][0];



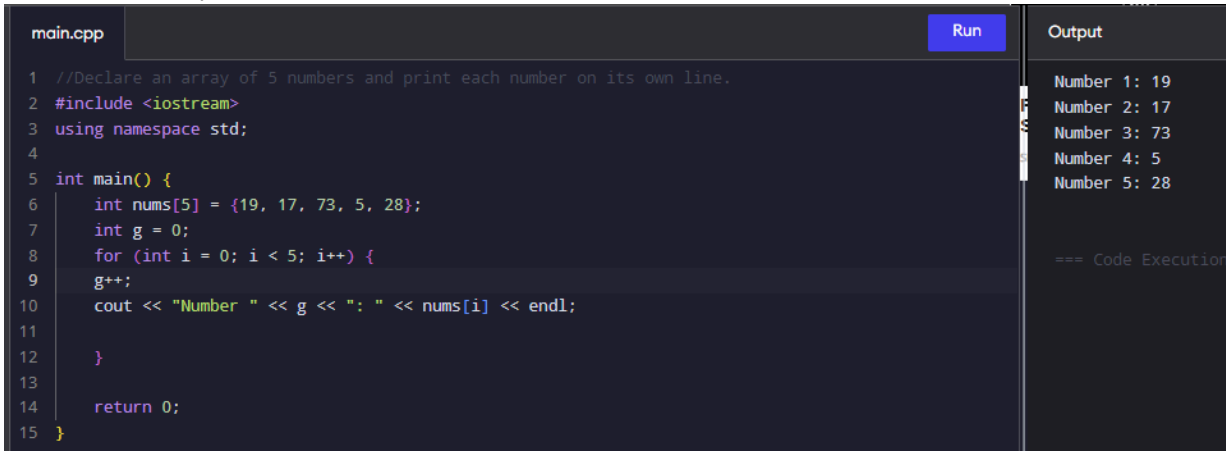
This code is like a nested array wherein there is a group of data inside a group of data. When outputting the desired value, the program read the code where in the 1 stands for the second group of data and the 0 is the first number in that chosen group of data.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

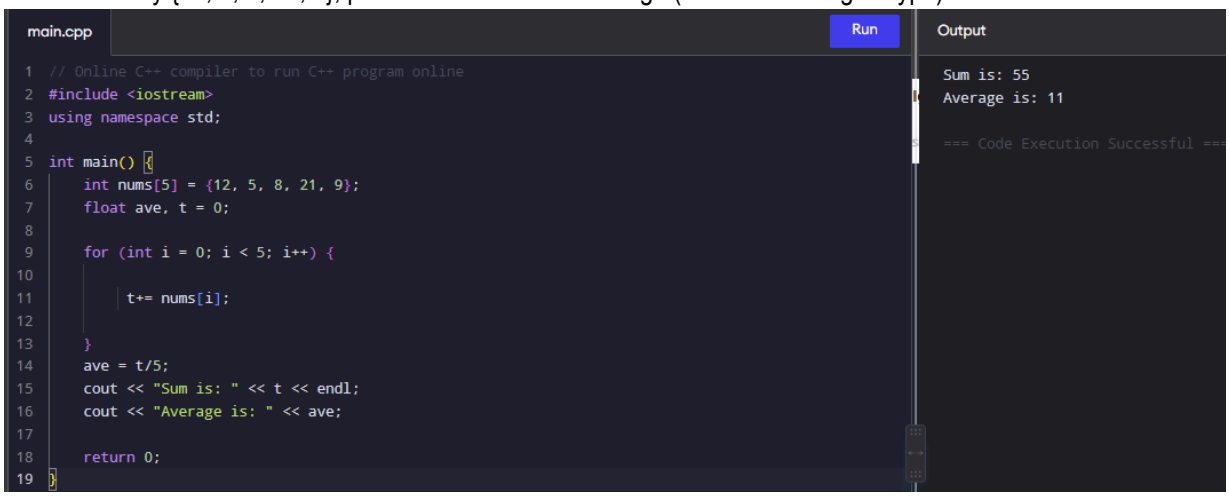
II. Programming exercise

1. Declare an array of 5 numbers and print each number on its own line.



This program checks each number located in the array which then prints it out in its own line with the help of the end line function. I added the variable g to represent their positions in order realistically, since in coding terms, the first position starts from 0.

2. For the array {12, 5, 8, 21, 9}, print the sum and the average (mind the average's type).



This loop program uses an array that has a group of numbers that will be added together as the for loop determines the position of each data in the array. For the average, since an array is definite and has the exact number of data inside it, I used that part as the divider for the total sum that the program has collected. Lastly is to output the sum and average of the numbers in the array.

3. Find and print the largest AND smallest number in an array.

main.cpp

```cpp
//Find and print the largest AND smallest number in an array.
#include <iostream>
using namespace std;

int main() {
    const int size = 5;
    int nums[size] = {5, 7, 8, 4, 15};
    int biggest = nums[0];
    int smallest = nums[0];

    for (int i = 0; i < size; i++) {

        if (nums[i] > biggest) {
            biggest = nums[i];
        }
        else if (nums[i] < smallest ) {
            smallest = nums[i];
        }
    }
    cout << "The biggest number is: " << biggest << endl;
    cout << "The smallest number is: " << smallest;

    return 0;
}
```

Output

```
The biggest number is: 15
The smallest number is: 4

=== Code Execution Successful =
```

During our discussion, we encountered the same situation, however, only the biggest was asked to be located. Here in this problem, I simply applied the same principle on how to get the biggest but this time in the concept of getting the smallest which will then be printed as the program or loop terminates.

4. Count how many numbers in an array are even.

main.cpp

```cpp
//Count how many numbers in an array are even.
#include <iostream>
using namespace std;

int main() {
    const int size = 5;
    int nums[size] = {5, 7, 8, 4, 15};
    int g = 0;

    for (int i = 0; i < size; i++) {

        if (nums[i] % 2 == 0) {
            g+= 1;
        }

    }
    cout << "There are " << g << " even numbers in the array.";

    return 0;
}
```

Output

```
There are 2 even numbers in the array.

=== Code Execution Successful ===
```

This program follows the principle that even numbers are divisible by 2, hence, I used the modulo function that collects the remainder of a division process. If the remainder is 0 therefore it is an even number, to acquire its number of even, I declared a variable for its very own counter.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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